GRADE 7 ACCELERATED · UNIT 5 · ASSESSMENT

Multi-Step Proportional Relationships

Form A · 24 points **WORKED ANSWERS**

Show your work. One point for each lettered part.

Part 1. Scale Drawings and Scale Factors *10 points*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

1. A 4 cm side of a figure is 10 cm in its scale copy. Find the scale factor. *1 pt*

k = 2.5

copy ÷ original

Scoring: 1 point.

2. A 15 cm side of a figure is 6 cm in its scale copy. Find the scale factor. *1 pt*

k = 0.4

copy ÷ original; the copy is smaller, so k is under 1 — 2.5 is the proportion backwards

Scoring: 1 point.

3. A figure is scaled by a scale factor of 3. Select ALL of the statements that are true about the copy. *1 pt*

- ☒ **A. Each side is 3 times as long.**
- ☐ B. Each angle is 3 times as large.
- ☒ **C. The perimeter is 3 times as long.**
- ☐ D. The area is 3 times as large.
- ☒ **E. The area is 9 times as large.**

A, C and E

All three and nothing else for the point: sides and perimeter × k, area × k^2, angles unchanged.

Scoring: 1 point.

4. A 3 cm by 4 cm rectangle is scaled by k = 3.

a. Find the copy's perimeter. *1 pt*

42 cm

perimeter × k

b. Find the copy's area. *1 pt*

108 cm^2

area × k^2 — 36 multiplied the area by k once

Scoring: a 1 · b 1 = 2 points.

5. A figure has area 8 cm^2. Its scale copy has area 72 cm^2. What is the scale factor? *1 pt*

k = 3

72 ÷ 8 = 9 is k^2, so k = √9. A board with 9 handed in the area ratio as k.

Scoring: 1 point.

6. A floor plan is drawn at ¼ in : 1 ft. A room is 3 in by 3.5 in on the plan.

a. Find the real length of the wall drawn 3 in long. *1 pt*

12 ft

the scale per 1 inch, then ×: 1 in : 4 ft

b. Find the room's real area. *1 pt*

168 ft^2

real dimensions first: 12 by 14 — 10.5 in^2 × 4 = 42 is the named error

Scoring: a 1 · b 1 = 2 points.

7. A plan is drawn at 1 in : 4 ft. A real wall is 18 ft long. How long is it on the plan? *1 pt*

4.5 in

real ÷ rate

Scoring: 1 point.

8. A plan is drawn at 1 in : 5 ft. A room is 2 in by 2 in on the plan. 1 ft = 0.305 m. Find the room's real area in square meters. *1 pt*

About 9.30 m^2 (3.05 m by 3.05 m); 9.29 by the sheet's other ratio

10 ft by 10 ft; 3.05 × 3.05 = 9.3025. The same number by the other route: 100 ft^2 × 0.305^2 = 9.3025; by the sheet's 1 m = 3.28 ft, 100 ÷ 3.28^2 = 9.29 — full credit. 30.5 (100 × 0.305) used the ratio once on an area. A plan's area asked in metric appears on no review and in no question bank; its type was 5.10 board 8.

Scoring: 1 point.

Part 2. Converting Across Systems *8 points*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

9. 1 in = 2.54 cm. How many centimeters is 9 in? *1 pt*

22.86 cm (23.1 by ÷ 0.39)

× the ratio whose 1 is inches; the sheet's other ratio, ÷ 0.39, earns the point too when the work shows it

Scoring: 1 point.

10. 1 in = 2.54 cm and 1 cm = 0.39 in. Select ALL of the expressions that give the number of inches in 70 cm. *1 pt*

- ☒ **A. 70 × 0.39**
- ☐ B. 70 × 2.54
- ☒ **C. 70 ÷ 2.54**
- ☐ D. 70 ÷ 0.39

A and C

Both sheet routes (27.3 and 27.6) and nothing else for the point.

Scoring: 1 point.

11. 1 yd = 3 ft, 1 ft = 12 in, 1 in = 2.54 cm. How many centimeters is 4 yd? *1 pt*

365.76 cm

a chain: 144 in, then × 2.54 (144 ÷ 0.39 = 369.2 by the other ratio — full credit). The estimate (a yard is nearly a meter) catches 10.16.

Scoring: 1 point.

12. 1 kg = 2.204 lb. How many pounds is 20 kg? *1 pt*

44.08 lb (44.05 by ÷ 0.454)

× the ratio whose 1 is the unit you have; the sheet's other ratio earns the point too when the work shows it

Scoring: 1 point.

13. 1 gal = 3.785 L. How many liters is 5 gal? *1 pt*

18.925 L (18.94 by ÷ 0.264)

more liters than gallons; the sheet's other ratio earns the point too when the work shows it

Scoring: 1 point.

14. 1 ft = 0.305 m. A patio is 20 ft^2 in area. Which is closest to its area in square meters? *1 pt*

- ○ A. 6.1 m^2
- ○ B. 0.093 m^2
- ● **C. 1.86 m^2**
- ○ D. 12.2 m^2

C

20 × 0.305² = 1.8605 (or 20 ÷ 3.28² = 1.86).

Scoring: 1 point.

15. Rate on Nov 5: 1 US dollar = 0.92 euros.

a. How many euros is $60? *1 pt*

55.20 euros

× the rate

b. How many US dollars is 23 euros? *1 pt*

$25

÷ the rate; 21.16 multiplied both ways

Scoring: a 1 · b 1 = 2 points.

Part 3. Multi-Step Proportion Problems *6 points*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

16. A car gets 26 miles per gallon and its tank holds 15 gal.

a. How far does it go on a full tank? *1 pt*

390 mi

the rate × the gallons

b. It makes a 1,000-mi trip. How many tanks of gas does the trip use? (Part of a tank counts as a whole tank.) *1 pt*

3 tanks (1,000 ÷ 390 ≈ 2.56)

the bracket on the paper settles the rounding; follow-through on part (a)

Scoring: a 1 · b 1 = 2 points.

17. A recipe for 4 servings uses $1\frac{1}{2}$ cups of rice.

a. How much rice for 10 servings? *1 pt*

3¾ cups

k = 2.5 from the yields, then the ingredient

b. With 6 cups on hand, how many servings can be made? *1 pt*

16 servings

4 recipes × the yield; 24 scaled the yield by the cups

Scoring: a 1 · b 1 = 2 points.

18. Juice is mixed orange : mango : water = 3 : 2 : 4. A batch uses 9 cups of orange. How many cups is the whole batch? *1 pt*

27 cups

k = 3 from the orange, then the whole, 9 parts; 6 found the mango and stopped

Scoring: 1 point.

19. 1 mi = 1.61 km. A drive is 200 km long. Car A averages 40 mi/h; car B averages 60 km/h. Which car gets there first, and by about how many minutes? Show both speeds in one unit. *1 pt*

Car A, by about 14 minutes: 40 mi/h = 64.4 km/h, so 200 ÷ 64.4 ≈ 3.11 h against 200 ÷ 60 ≈ 3.33 h.

The point needs the speeds in one unit and the choice with its gap: 0.227 h is 13.7 min — 13 or 14 both earn it (by the sheet's 0.62 route, 124 mi: 3.1 h against 3.33 h, 14 min). 'Car B, 60 > 40' compared two units. A decision from two rates in different units appears on no review and in no question bank; its type was 5.10 board 9.

Scoring: 1 point.